

TrES-3b

R-1969

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_131008 · created 2026-08-06T23:45:18+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456573.6179 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.058
Transit depth (Rp/Rs)^2	3.7 +/- 2.21 [%]
Orbital Inclination (inc)	85.86 +/- 2.92
Airmass coefficient 1 (a1)	2.4 +/- 0.38
Airmass coefficient 2 (a2)	-0.63 +/- 0.18
Scatter in the residuals of the lightcurve fit is	16.02 %
Best Comparison Star	#2 - [34, 302]
Optimal Aperture	1.4
Optimal Annulus	9.28
Transit Duration (day)	0.0574 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.058 vs 0.16309 (deviation 0.5σ)
Duration fit vs published	0.0574 d vs 0.0567 d (deviation 0.09σ)
Mid-time O–C	-3.2 min
Depth SNR	3.3
Residual scatter	16.02 %

Light curve

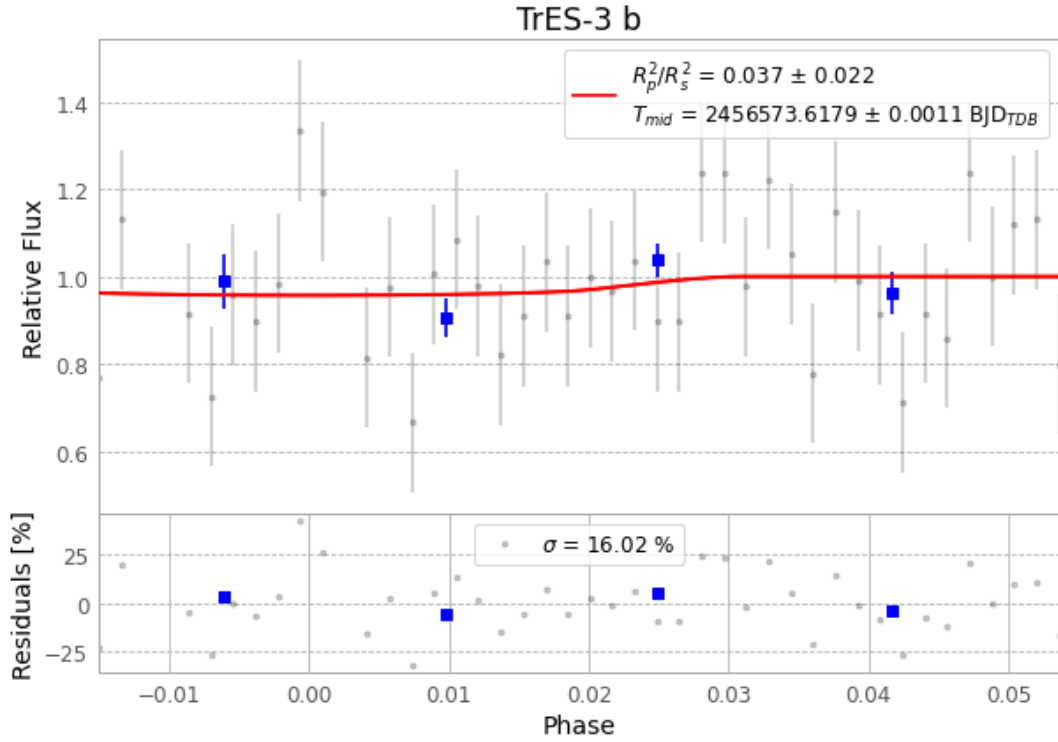


Figure 1 — FinalLightCurve_TrES-3 b_2013-10-07.png (SHA-256 f6375daf5ec536b9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [369, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6a1716d5fe08f368...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

44 FITS frames (29 MB), TRES-3131008022112.FITS → TRES-3131008043012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-131008.log (6a968f9acf2f...), FinalLightCurve_TrES-3 b_2013-10-07.png (f6375daf5ec5...), FinalLightCurve_TrES-3 b_2013-10-07.csv (dfe404b3d730...)

Compute resources

Wall time 117.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 23:45Z.